UNSUPERVISED LEARNING

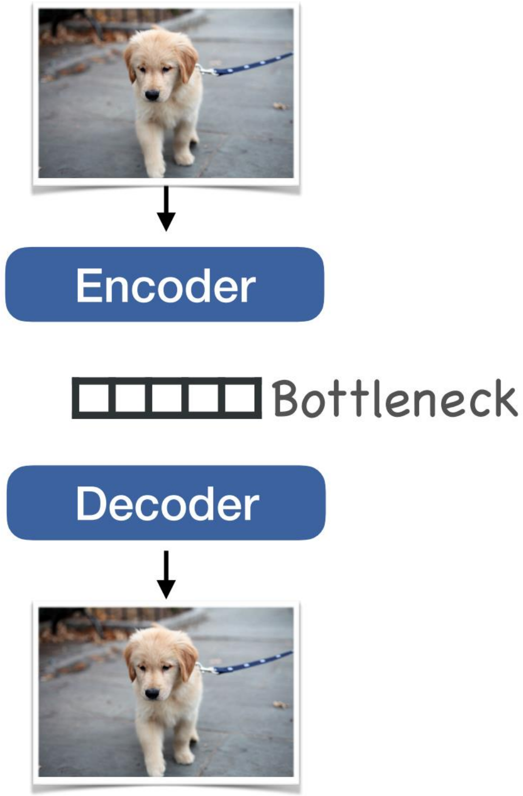
OSCAR LLORENTE GONZALEZ

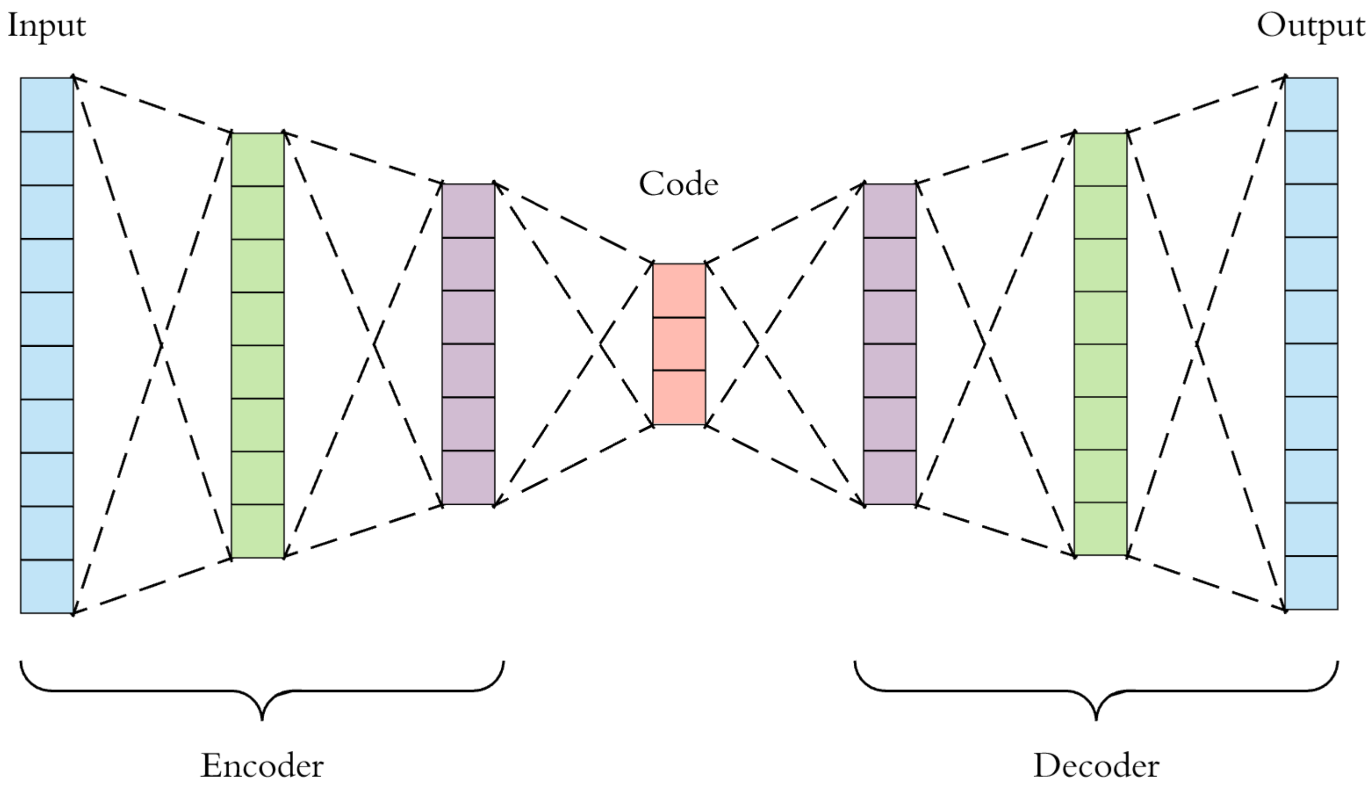
OLLORENTE@COMILLAS.EDU

INDEX

1 - INTRODUCTION

WHAT IS AN AUTOENCODER?





HOW TO TRAIN AUTOENCODER

$$p_{\text{encoder}}(\boldsymbol{h} \mid \boldsymbol{x})$$

$$p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h})$$

$$L(\boldsymbol{x}, g(f(\boldsymbol{x})))$$

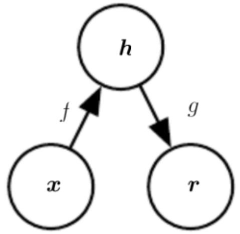


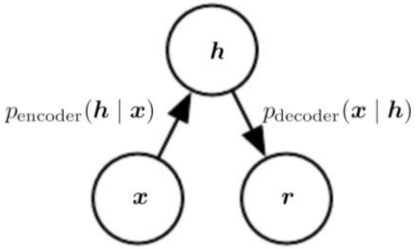
Figure 14.1: The general structure of an autoencoder, mapping an input $\boldsymbol{x}$ to an output (called reconstruction) $\boldsymbol{r}$ through an internal representation or code $\boldsymbol{h}$. The autoencoder has two components: the encoder f (mapping $\boldsymbol{x}$ to $\boldsymbol{h}$) and the decoder g (mapping $\boldsymbol{h}$ to $\boldsymbol{r}$).

STOCHASTIC ENCODERS AND DECODERS

$$p_{\text{encoder}}(\boldsymbol{h} \mid \boldsymbol{x}) = p_{\text{model}}(\boldsymbol{h} \mid \boldsymbol{x})$$

$$p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h}) = p_{\text{model}}(\boldsymbol{x} \mid \boldsymbol{h}).$$

$$-\log p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h})$$

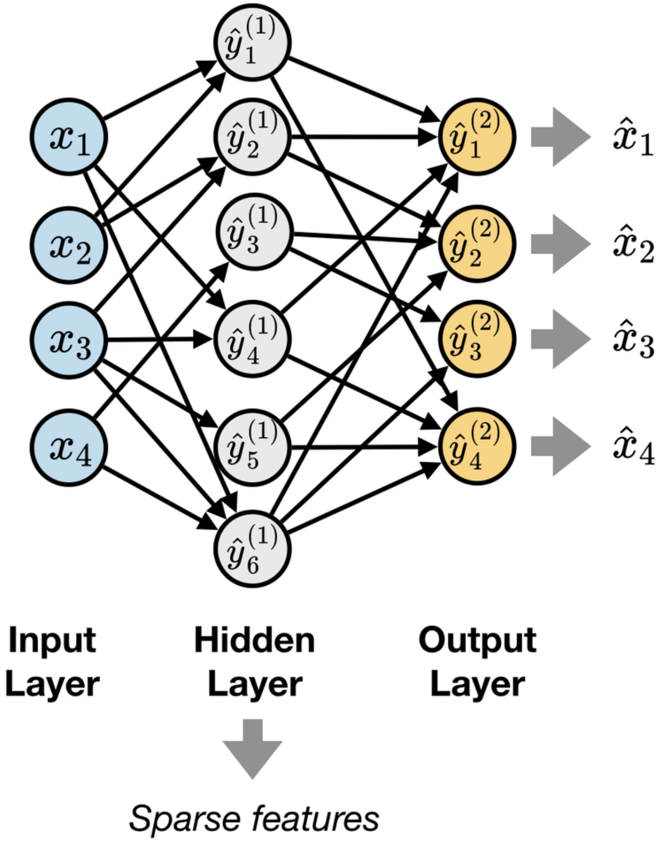


2 - SPARSE AUTOENCODERS

SPARSE AUTOENCODERS

$$L(\boldsymbol{x}, g(f(\boldsymbol{x}))) + \Omega(\boldsymbol{h}),$$

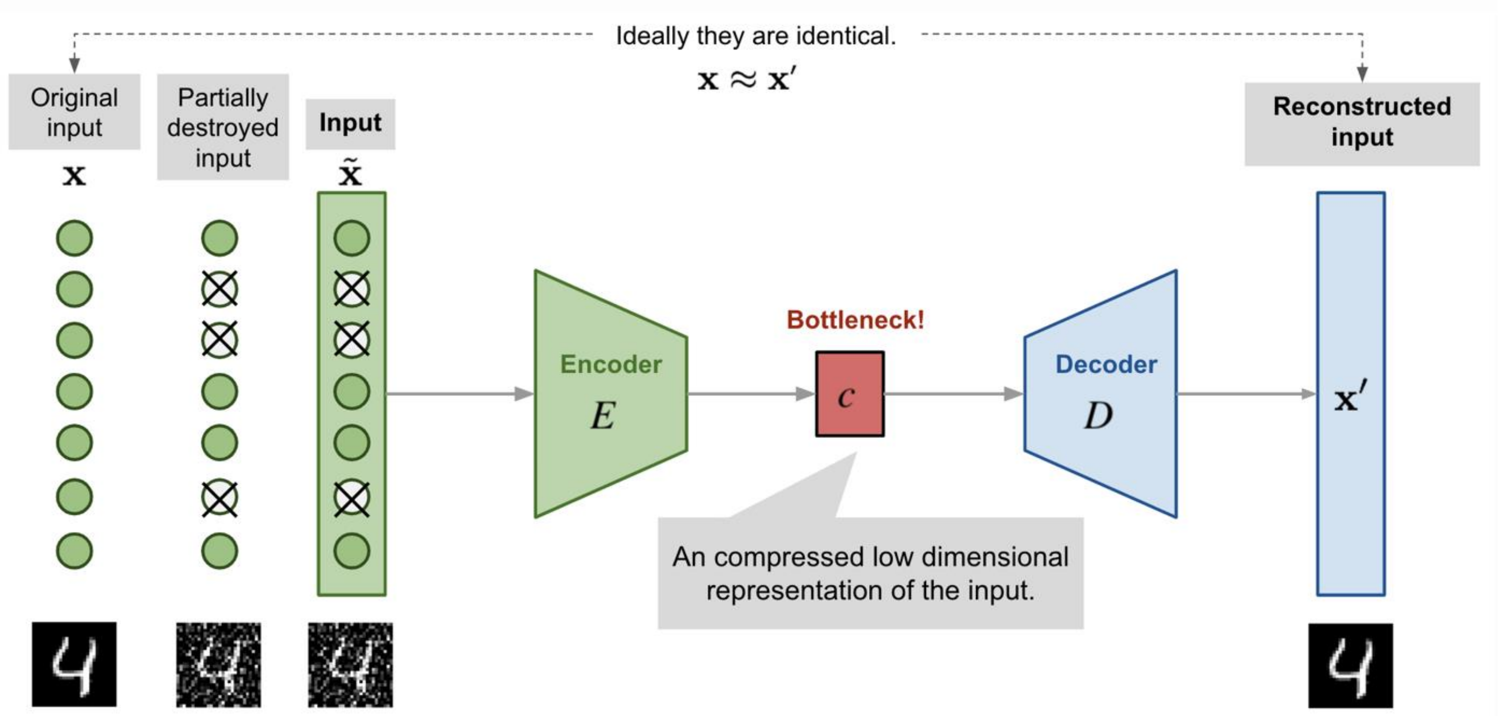
$$\Omega(\boldsymbol{h}) = \lambda \sum_i |h_i|,$$



3 - DENOISING AUTOENCODERS (DAE)

INTRODUCTION TO DAE

$$L(\boldsymbol{x}, g(f(\tilde{\boldsymbol{x}}))),$$



STRUCTURE OF DAE

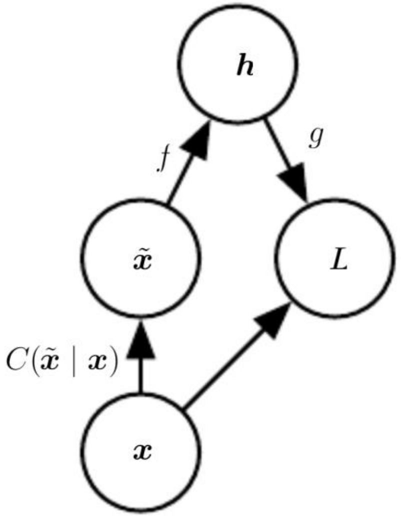


Figure 14.3: The computational graph of the cost function for a denoising autoencoder, which is trained to reconstruct the clean data point $\boldsymbol{x}$ from its corrupted version $\tilde{\boldsymbol{x}}$. This is accomplished by minimizing the loss $L = -\log p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h} = f(\tilde{\boldsymbol{x}}))$, where $\tilde{\boldsymbol{x}}$ is a corrupted version of the data example $\boldsymbol{x}$, obtained through a given corruption process $C(\tilde{\boldsymbol{x}} \mid \boldsymbol{x})$. Typically the distribution p_{decoder} is a factorial distribution whose mean parameters are emitted by a feedforward network g.

HOW TO TRAIN A DAE

1. Sample a training example $\boldsymbol{x}$ from the training data.

2. Sample a corrupted version $\tilde{\boldsymbol{x}}$ from $C(\tilde{\mathbf{x}} \mid \mathbf{x} = \boldsymbol{x})$.

3. Use $(\boldsymbol{x}, \tilde{\boldsymbol{x}})$ as a training example for estimating the autoencoder reconstruction distribution $p_{\text{reconstruct}}(\boldsymbol{x} \mid \tilde{\boldsymbol{x}}) = p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h})$ with $\boldsymbol{h}$ the output of encoder $f(\tilde{\boldsymbol{x}})$ and p_{decoder} typically defined by a decoder $g(\boldsymbol{h})$.

$$-\mathbb{E}_{\mathbf{x} \sim \hat{p}_{\text{data}}(\mathbf{x})} \mathbb{E}_{\tilde{\mathbf{x}} \sim C(\tilde{\mathbf{x}} \mid \boldsymbol{x})} \log p_{\text{decoder}}(\boldsymbol{x} \mid \boldsymbol{h} = f(\tilde{\boldsymbol{x}})),$$

4 - CONTRACTIVE AUTOENCODERS (CAE)

INTRODUCTION TO CONTRACTIVE AUTOENCODERS

$$L(\boldsymbol{x}, g(f(\boldsymbol{x}))) + \Omega(\boldsymbol{h}, \boldsymbol{x}),$$

$$\Omega(\boldsymbol{h}, \boldsymbol{x}) = \lambda \sum_i ||\nabla_{\boldsymbol{x}} h_i||^2.$$

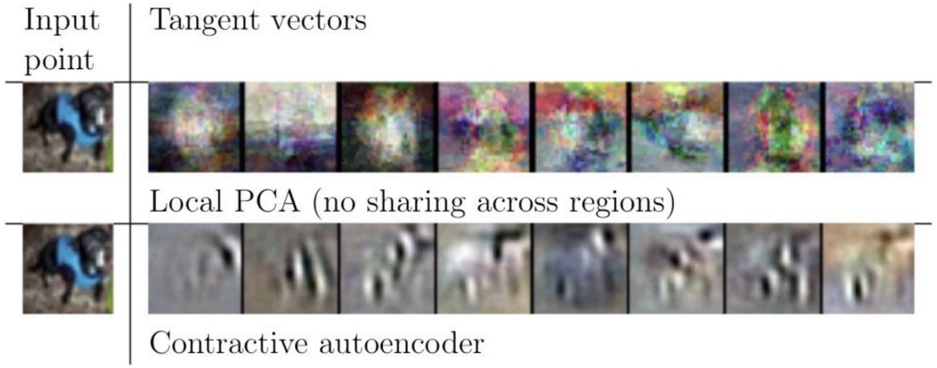


Figure 14.10: Illustration of tangent vectors of the manifold estimated by local PCA and by a contractive autoencoder. The location on the manifold is defined by the input image of a dog drawn from the CIFAR-10 dataset. The tangent vectors are estimated by the leading singular vectors of the Jacobian matrix $\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{x}}$ of the input-to-code mapping. Although both local PCA and the CAE can capture local tangents, the CAE is able to form more accurate estimates from limited training data because it exploits parameter sharing across different locations that share a subset of active hidden units. The CAE tangent directions typically correspond to moving or changing parts of the object (such as the head or legs). Images reproduced with permission from Rifai *et al.* (2011c).

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